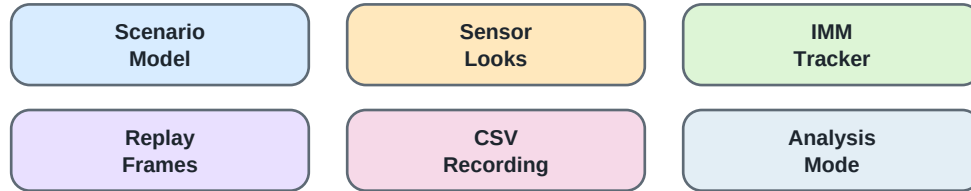


ECEF Target Tracker

Repository and algorithm overview for the Java/Swing target-scenario, tracking, recording, replay, and track-stitching analysis application.



What this application does

The project is a Java 17 Swing desktop application for building and replaying target motion scenarios on a Plate Carree Earth view. User-created and pre-generated target paths are stored in ECEF meters, while path construction and altitude handling respect WGS-84 surface geometry. A configurable omniscient sensor creates noisy ECEF position and velocity measurements, an IMM tracker estimates target tracks, and Analysis Mode replays saved scenario runs for track-stitching studies.

- Scenario Generation Mode: edit targets, presets, profiles, sensor settings, blackout regions, IMM settings, recording, pre-compute, and visual replay.
- Analysis Mode: load a recorded run, replay tracks/truth/measurements, and run embedded Track Stitching Analysis.
- No runtime internet dependency: map detail uses bundled raster/vector resources.

Package map

Package	Responsibility	Representative classes
com.targettracker.model	Scenario state, WGS-84/ECEF math, target paths, profiles, blackout regions, presets, saved scenario definitions.	ScenarioModel, TargetTrajectory, Wgs84, Wgs84Geodesic, PresetScenarioGenerator
com.targettracker.ui	Swing application frame, map canvas, sidebar cards, pre-compute/replay loop, sensor simulation, analysis panels.	TrackerFrame, EarthMapCanvas, ScenarioPlayback, MeasurementEngine, TrackStitchingAnalysisPanel
com.targettracker.tracking	IMM filter, track views, track records, association support, linear algebra.	ImmTracker, ImmModel, ImmSettings, TrackRecord, TrackView
com.targettracker.recording	MATLAB-friendly CSV output and ingestion for recorded scenarios.	TrackCsvRecorder, TrackCsvReader, RecordedScenario
com.targettracker.analysis	Track stitching timing/cost analysis and full diagnostic export.	TrackStitchingAnalyzer, TrackStitchingAnalysisExporter

Scenario and UI workflow

The main frame keeps one world view and one replaceable right-side control card. Top-level mode controls switch between Scenario Generation and Analysis Mode. The timeline under the map becomes seekable after pre-computation or after loading a recorded scenario.

UI area	Current behavior
World view	Plate Carree map with zoom/pan, latitude/longitude axes, local scale cues, bundled geographic detail, target truth, measurements, track means, tails, covariance ellipses, blackout regions, and stitch-analysis focus modes.
Scenario card	Preset selector, origin latitude/longitude, average speed, altitude, duration, user-scenario save controls.
Targets card	Target selector, add/remove target, drawing controls, clear/finish path, velocity profile, altitude profile, and path-prefix highlighting while profile cursor moves.
Sensor Parameters card	Look interval/offset, position and velocity standard deviations, probability of detection, and user-drawn blackout rectangles.
IMM card	CV/CA model toggles, process noise entries, association threshold, break thresholds, and transition matrix.
Analysis Mode card	Recorded-scenario folder loader, layer toggles, history sliders, and Track Stitching Analysis entry point.

Scenario pre-computation and replay

Pre-computation runs the full scenario as fast as possible at 0.1-second steps. Each step records ground truth, advances measurement generation, updates the IMM at measurement times, captures replay frames, and optionally writes CSV output. Replay is video-like: it displays cached frames and supports pause, reset, and timeline scrubbing without rerunning the tracker.

Coordinate and motion model

- Target control points and animated states are represented in ECEF.
- Geodetic interpolation follows WGS-84 ellipsoidal geometry rather than straight Cartesian chords.
- Altitude profiles are applied as WGS-84 ellipsoidal height before conversion back to ECEF.
- Velocity and altitude profile editors operate over normalized path time and visually thicken the corresponding path prefix.

Sensor and blackout model

- The initial sensor is a God sensor: every target is considered at scheduled look times.
- Look times are offset + $n \cdot \text{interval}$.
- Each target receives an independent probability-of-detection coin flip.
- Successful measurements add independent zero-mean Gaussian noise to 3D ECEF position and 3D ECEF velocity.
- Blackout regions suppress measurement creation while the target is inside the region.
- User blackout regions are two-click rectangles drawn on the map; internal IDs are stored but not displayed.

IMM tracker

The tracker is an ECEF interacting multiple model estimator. CV and CA models are enabled by default, and each track holds a fused 9D state ordered as $[x, y, z, vx, vy, vz, ax, ay, az]$. The UI predicts views between updates, but the filter state is only committed at measurement updates so coast drawing does not feed back into the covariance.

Part	Algorithm
Models	CV uses discretized continuous white-noise acceleration. CA uses the analogous continuous white-jerk process over position, velocity, and acceleration.
Association	Gated track/measurement pairs are scored by Mahalanobis distance. A greedy global assignment gives each measurement and each track at most one match.
Births	Unassigned measurements can spawn new tracks.
Break logic	Tracks with no update past timeout or with excessive 3D covariance radius are marked dead. Dead tracks stay visible in grey.
Recording	Integer-second snapshots are written for all tracks, plus fractional rows at measurement updates when needed.

Recording and Analysis Mode

Recorded runs are stored as UTF-8 CSV in `ground_truth_data`, `track_data`, and `measurement_data` directories. Track records include 9D state, 9x9 covariance, and updated/coasted status. Measurements include sensor ID, source target ID, associated track ID, 6D mean, 6x6 covariance, and uncertainty values. Analysis Mode reconstructs replay from those files.

Track Stitching Analysis

Track Stitching Analysis searches measurement times where at least one old/coasting track and one recently formed distinct track are both eligible. For each old/new pair it estimates join time, evaluates compatibility, and computes optimal assignments by Hungarian solve.

Step	Current implementation
Candidate selection	Old tracks satisfy the coasted-track window. New tracks satisfy the new-track-age window. Dead old tracks may be included when enabled.
Join-time estimates	Simple midpoint, kinematic midpoint, Mahalanobis time-bank midpoint, and truth-only RMS reference time.
Propagation	Old tracks predict forward from last update. New tracks retrodict from the event snapshot through earlier associated measurements, then continue backward if needed.
Innovation	Full 9D innovation is $x_{old} - x_{new}$. Innovation covariance is $P_{old} + P_{new}$.
Feasibility costs	Canonical Gaussian negative log likelihood and canonical Mahalanobis distance.
Alternative hypothesis costs	NLLR against static/uniform extraneous density and learned spatial birth intensity.
Assignment	Hungarian assignment is run separately for NLL, Mahalanobis, static NLLR, and learned NLLR. Truth RMS is excluded from optimal assignment.

Learned spatial birth intensity

The learned null model is now a local Gaussian-mixture intensity over track births, seeded only by the first associated measurement for each track. This avoids diluting dense birth sites, such as hangars or airports, across the full operating volume.

- One birth component is created per track, using that track's first associated measurement position and covariance.
- The component is activated online when its first measurement time has occurred.
- Each component is a normalized 3D Gaussian intensity in births per cubic kilometer.
- A minimum kernel width keeps nearby births mutually reinforcing instead of creating isolated spikes.
- Component weights are downweighted if the birth is spatially compatible with a previous track continuation.
- Time forgetting reduces old component weights using a configurable half-life.
- The NLLR query evaluates local expected unrelated births by integrating density over the innovation volume, approximated as local intensity times volume with uncertainty smoothing.

Operationally, a cluster of first detections around an airport should produce higher learned birth intensity than a single isolated hard-turn scenario. The smoke tests now include a regression check that clustered first-measurement births produce a much higher representative intensity than a single isolated birth.

Track Stitching Analysis export

The Track Stitching Analysis panel includes an output parent-folder text box and an Output data to folder button. Exports create a scenario/timestamp folder with summary files, full time-bank diagnostics, and density history.

Export path	Contents
README.md	How to read the export and MATLAB readable examples.
summary/configuration.csv	Scenario metadata and analysis settings.
summary/events.csv	Candidate timestamp counts and representative learned density.
summary/segments.csv	All segment anchors, states, covariances, candidate flags, and live/dead state.
summary/pair_time_estimates.csv	All join-time estimates and NLL/Mahalanobis/NLLR values.
summary/optimal_assignments.csv	Hungarian-solver choices by metric.
bank_evaluations/bank_evaluations.csv	Old/new state and covariance, innovation and innovation covariance, Mahalanobis, NLL, static NLLR, learned NLLR for each bank time.
spatial_density/spatial_density_history.csv	Representative/peak learned birth intensity, active component count, total component weight, kernel size, and operating bounds over time.

Important implementation notes

- The app is intentionally dependency-light at runtime; Maven builds Java 17 code with no third-party Java libraries.
- Most test coverage is smoke-style and executable as Java main classes under src/test/java.
- The map projection is Plate Carree for display only; trajectory math uses WGS-84 geodesic geometry and ECEF state storage.
- Analysis Mode never reruns the tracker; it replays data loaded from recording CSVs.
- The learned spatial density is intended as an operational null hypothesis for extraneous track births, not a truth-label classifier.